# DEEP FAKE FACE DETECTION USING MACHINE LEARNING

# SDGs and TARGETs

- **Primary Goal No** : 16 (Peace, Justice and Strong Institution)
  - SDG TARGET1: Ensure public access to information and protect fundamental freedoms.
  - SDG TARGET2: Develop effective, accountable, and transparent institutions.
  - SDG TARGET3: Strengthen institutions to prevent violence and combat misinformation.
- **Secondary Goal No** : 4 (Quality Education)
  - SDG TARGET1: Increase skills for employment, decent jobs, and entrepreneurship.
  - SDG TARGET2: Promote knowledge and skills for sustainable development, including media literacy.
  - SDG TARGET3: Ensure equal access to affordable quality technical education and training
- **Tertiary Goal No** : 9 (Industry, Innovation and Infrastructure)
  - SDG TARGET1: Enhance scientific research and upgrade technological capabilities.
  - SDG TARGET2: Increase access to information and communication technology.
  - SDG TARGET3: Develop quality, reliable, sustainable infrastructure.

# Problem Statement

The rapid advancement of deepfake technology has enabled the creation of highly realistic fake face images and videos using machine learning techniques. These manipulated media are increasingly difficult for humans and traditional detection systems to identify, leading to serious issues such as misinformation, identity misuse, reputational damage, psychological harm, and loss of trust in digital media. Existing detection approaches often lack robustness and fail to generalize across different deepfake techniques and real-world scenarios. Therefore, there is a strong need for an automated and reliable deepfake face detection system that can accurately distinguish real and fake content by analyzing facial features and patterns using machine learning techniques, thereby enhancing digital safety and trust.The proposed platform provides an accessible, efficient, and reliable tool for deepfake detection.

# Abstract

Deepfake technology has rapidly evolved, enabling the creation of highly realistic fake images and videos that are difficult to distinguish from genuine media. The widespread use of such manipulated content poses serious risks, including misinformation, identity misuse, and reputational damage. Traditional detection methods often fail to identify these advanced manipulations effectively. To address this challenge, this project proposes a web-based deepfake detection platform for verifying media authenticity. Users can upload images or videos to the system for analysis. The proposed approach utilizes Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) models to analyze sequential and temporal facial features. By examining frame-level inconsistencies and temporal patterns, the system detects subtle manipulations with high accuracy. The platform provides an accessible and user-friendly solution for deepfake detection.

# Existing System and its Disadvantages

**1. CNN-Based Detection Systems:**Most existing deepfake detection systems use Convolutional Neural Networks (CNNs) to analyze spatial features such as facial textures, edges, and visual artifacts in images and video frames. These systems are effective in identifying manipulation at the frame level and offer good accuracy for image-based deepfake detection. However, they often fail to capture temporal inconsistencies across video frames.

**2. Hybrid CNN–LSTM Models:**To overcome this limitation, hybrid models combine CNNs with LSTM networks to analyze both spatial and temporal features. This approach improves detection accuracy by identifying motion and sequence-based artifacts. However, such systems require large datasets, high computational power, and are less suitable for real-time applications.

# Proposed System and its Advantages

- The proposed system is a web-based deepfake face detection platform that allows users to upload images or videos through a secure login interface. The input media is preprocessed by extracting facial regions and normalizing frames for analysis. A CNN extracts spatial facial features, which are analyzed by RNN and LSTM models to detect temporal inconsistencies. Based on this combined analysis, the system classifies the media as Real or Deepfake with a confidence score and stores the results securely in a database.
- The system accurately detects deepfake images and videos by analyzing both spatial and temporal features.Its web-based interface with secure login makes it easy and safe for users to verify media authenticity.The platform helps reduce misinformation and enhances digital trust and online safety.

# Research and Literature Review

**Paper 1:**

- **Title and Authors :** A Novel Approach for Detecting Deepfake Face Using Machine Learning Algorithms, M. Kumar, P. K. Rai, and P. Kumar (2024)
- **Algorithm Used:** Logistic Regression, SVM, Random Forest, KNN
- **Accuracy:** ~55%
- **Disadvantages:**Works mainly on images and fails to capture temporal features, making it less effective for video-based deepfake detection.

**Paper 2:**

- **Title and Authors:** DeepFake Detection Using Convolutional Neural Networks, Li, Zhang, and Wang (2023)
- **Algorithm Used:** CNN (VGG-19, ResNet-50)
- **Accuracy:** ~58%
- **Disadvantages:**Requires high computational resources.Not suitable for real-time applications

# Research and Literature Review

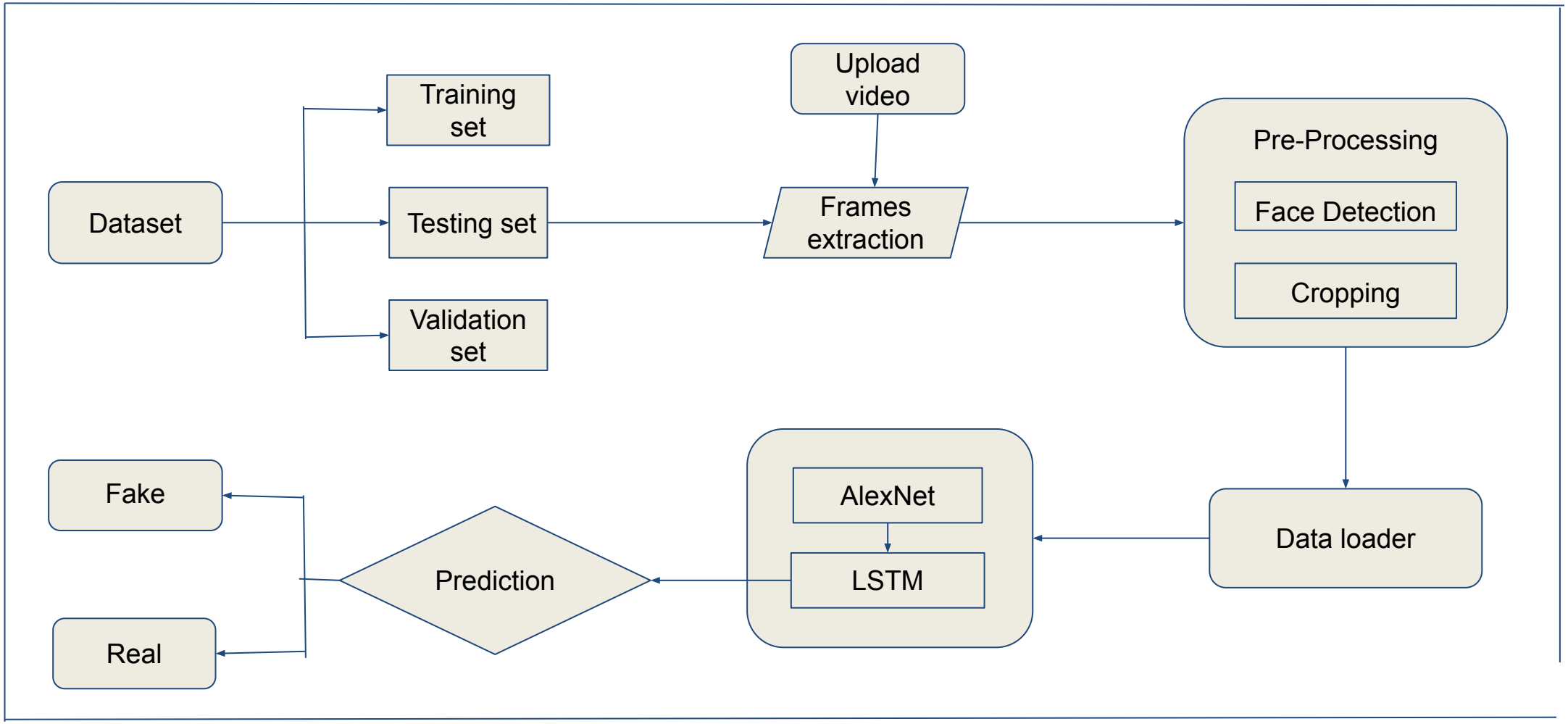
**Paper 3:**

- **Title and Authors:** Explainable AI for Deepfake Detection: A Human-Centric Approach, Y. Wang, L. Chen, and H. Zhao (2024)
- **Algorithm Used:** CNN with Explainable AI (XAI)
- **Accuracy:** ~52%
- **Disadvantages:**Added explainability increases complexity and reduces detection accuracy and speed.

**Paper 4:**

- **Title and Authors:** Video Deepfake Detection Using a Hybrid CNN-LSTM-Transformer Model for Identity Verification, Raghav Gupta, Meera Singh, Arun Sharma (2025)
- **Algorithm Used:** Hybrid CNN + LSTM + Transformer model (3DMM + deep learning)
- **Accuracy:** ~57%
- **Disadvantages:**The hybrid CNN-LSTM-Transformer model is computationally heav requires large GPU resources. It may overfit on some datasets.

# System Architecture



# Module Description

**1. Data Acquisition and Preprocessing**

Collect real and deep fake videos/images from reliable datasets such as FaceForensics++ and DFDC. The collected data is cleaned and preprocessed through face detection, cropping, resizing, normalization, and noise removal to create consistent input data. This step ensures that the dataset is properly prepared and suitable for effective model training.

**2. Feature Extraction and Representation**

Extract important spatial and temporal features from facial frames. The system analyzes facial expressions, skin texture, lighting, and frame-to-frame changes to identify patterns that may indicate deepfake manipulation. These features help the model understand the differences between authentic and manipulated facial conte

# Module Description

### 3. RNN-LSTM Model Architecture

Design a combined Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) model to process sequential video frames. RNN handles sequence information, while LSTM captures important short-term and long-term temporal dependencies.The combination helps the model effectively learn the changing patterns present across video frames.

### 4. Model Training and Validation

Train the RNN-LSTM model using the prepared dataset so that it learns the differences between real and fake content. The model is validated using unseen data to evaluate its performance and reduce overfitting.Performance is measured using metrics such as accuracy, precision, recall, and F1-score.

# Module Description

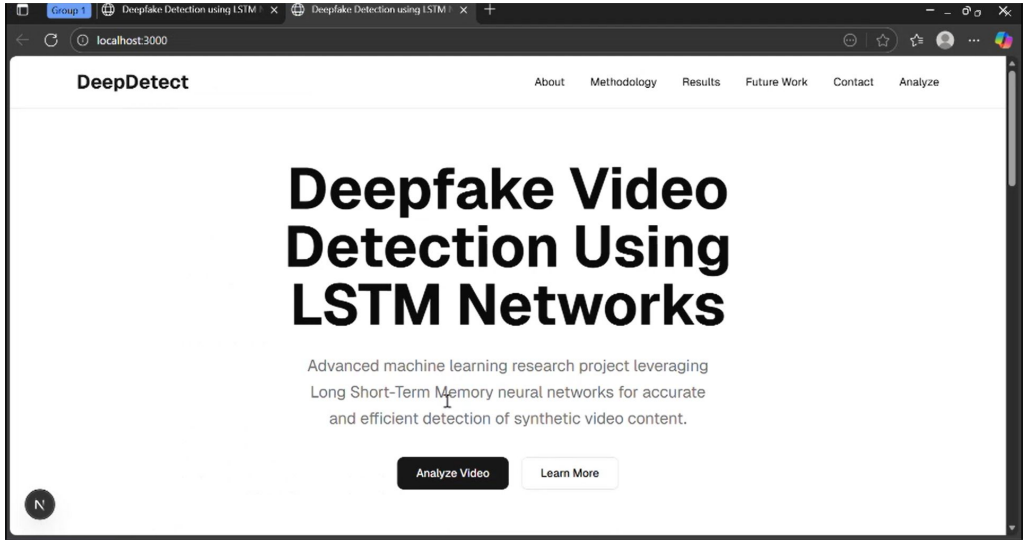
**5. Deepfake Detection and Classification**

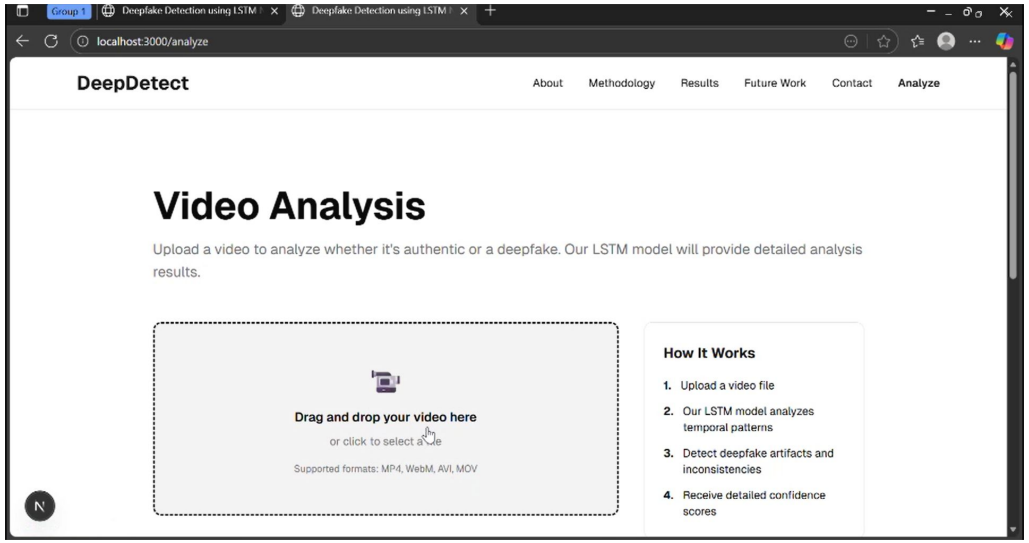
The trained model analyzes new uploaded videos or images and classifies them as Real or Fake. It can also provide a confidence or probability score indicating the reliability of the prediction. This module serves as the main component for identifying manipulated or authentic media content.

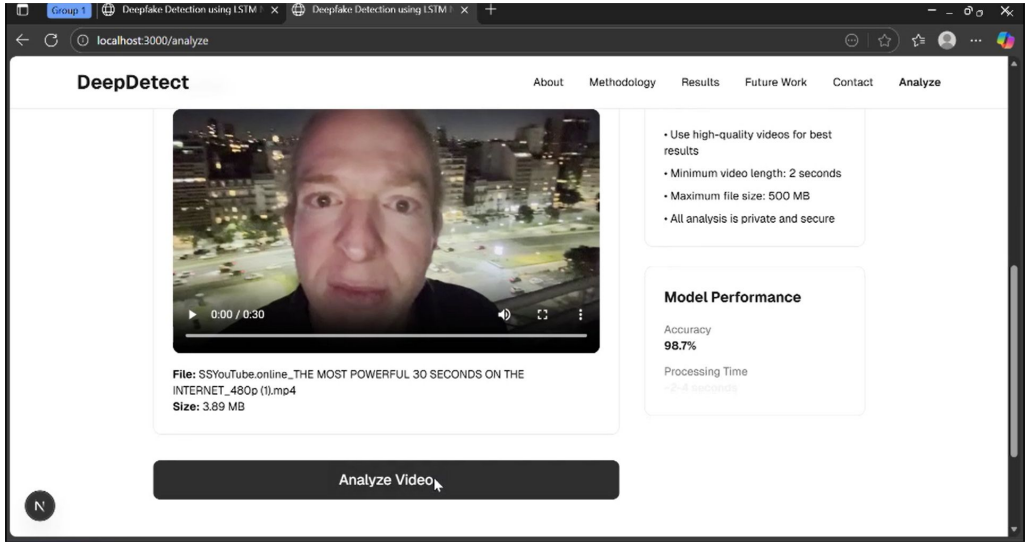
**6. Web Interface and Result Visualization**

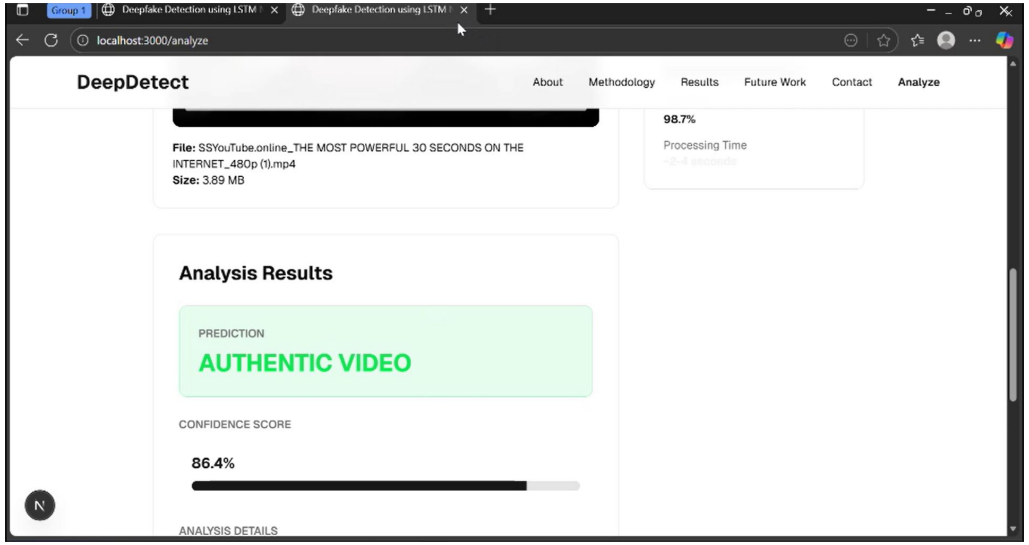
Develop a simple web-based interface that allows users to upload videos or images for analysis. The system processes the input using the trained model and displays the final detection result in an easy-to-understand format.The interface makes the deepfake detection system simple and accessible for non-technical users

# Experimental Results



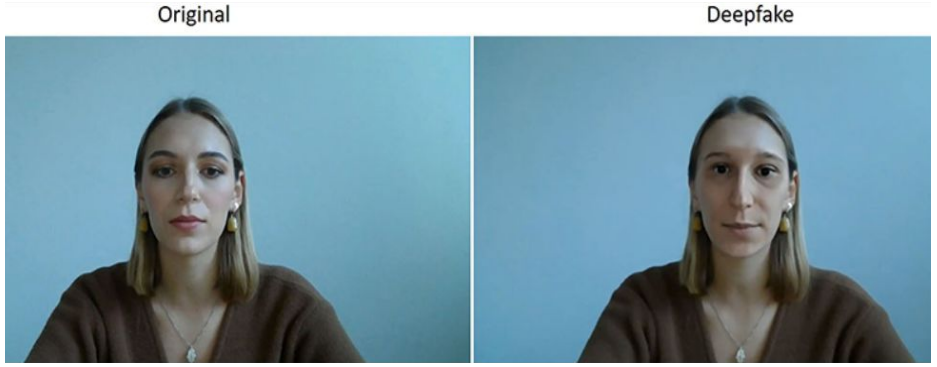



# Future Enhancements

- Use more and varied video data so the system can detect deep fakes more accurately in different conditions.
- Apply advanced deep learning models to identify very realistic deep fake videos.
- Improve the system to detect multiple faces in a single video and handle short videos better.
- Combine face movements and voice patterns to improve detection accuracy.
- Make the system faster for real-time detection on websites and mobile devices.
- Connect the system with social media and security platforms to automatically detect fake videos.





# Conclusion

Deepfake technology poses a growing threat by enabling the creation of highly realistic fake videos and images that can spread misinformation and harm reputations. Our project addresses this challenge by developing a web-based detection system using RNN and LSTM models, which analyze temporal and sequential features to accurately distinguish between real and manipulated media. By providing an accessible platform for users to verify content authenticity, this system contributes to combating the negative impact of deepfakes. Future improvements can focus on enhancing detection speed, expanding dataset diversity, and integrating more advanced deep learning techniques to keep pace with evolving deepfake methods. Additionally, the system can be extended for real-time detection and deployment on social media platforms. Overall, this project highlights the importance of responsible AI in ensuring digital safety and trust.

# Reference


- S. K. Singh, A. K. Singh, and R. K. Gupta, "DeepFake Detection using Inception ResNetV2 and LSTM," CEUR Workshop Proceedings, 2021.
- S. Agarwal and A. Verma, "Detecting Deepfakes With ResNeXt and LSTM," IEEE Xplore, 2023.
- M. V. Subba Rao, I. Bhargavi, A. Abhishek, and C. Phani Kiran, "Deep Fake Face Detection Using LSTM," Journal of Image Processing and Intelligent Remote Sensing, 2022.
- S. K. Singh, "Deepfake Detection Using Feature Extraction and LSTM," California State University, 2023.
- A. Sharma and R. K. Gupta, "Deepfake Detection Using Xception and LSTM," International Research Journal of Modernization in Engineering Technology and Science (IRJMETS), 2023.

# Reference


- S. Tariq, S. Lee, and S. S. Woo, “A Convolutional LSTM Based Residual Network for Deepfake Video Detection,” *arXiv preprint arXiv:2009.07480*, 2020.
- P. Saikia, D. Dholaria, P. Yadav, V. Patel, and M. Roy, “A Hybrid CNN-LSTM Model for Video Deepfake Detection by Leveraging Optical Flow Features,” *arXiv preprint arXiv:2208.00788*, 2022.
- J. Mallet, N. Krueger, M. Vanamala, and R. Dave, “Hybrid Deepfake Detection Utilizing MLP and LSTM,” *arXiv preprint arXiv:2304.14504*, 2023.
- O. A. H. H. Al-Dulaimi and S. Kurnaz, “An Investigation into the Utilisation of CNN with LSTM for Video Deepfake Detection,” *Applied Sciences*, 2024.
- S. K. Singh, A. K. Singh, and R. K. Gupta, “Deep Fake Detection using LSTM and Survey of Deep Fake Creation Techniques,” *International Journal of Intelligent Systems*, 2022.

# Thank You